

Retail Sales Prediction using Ensemble Machine learning Algorithm

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Abstract

Sales forecasting is a crucial component of supply chain management and retailer-manufacturer activities in the retail sector. The proliferation of digital data has reduced the use of traditional methods and systems for completing particular tasks. The most difficult responsibility for marketing, customer service, inventory management, and business financial planning in the retail sector is sales forecasting. In this study, we used various machine learning algorithms to undertake a predictive analysis of retail sales using the Kaggle database. To increase prediction accuracy, we used several machine learning techniques (SVM, XGBoost, Random Forest), models for sale forecasting, and thorough predictive analysis and assessment. There is still a big gap in managing complicated datasets with various product families and strong seasonality, notwithstanding developments. Using hyperparameter tuning via randomised search cross-validation, a RF model is implemented and optimised as part of the proposed solution. This strategy tackles the dataset's intricacies, identifying complicated patterns that conventional approaches overlook. Compared to the proposed model and standard RF, which had minor RMSE and MAE figures, the RSGDO-RFR model attained R-squared value of 0.94 and an Adjusted R-squared value of 0.95. The findings show that the RSGDO-RFR model performs exceptionally well in retail sales forecasting, managing the intricacy of the dataset with more accuracy and dependability. This study emphasize the value of sophisticated machine learning techniques in predictive analytics, providing a notable enhancement above conventional approaches and other modern models. This study aims to develop a more accurate prediction model with machine learning models. It has been demonstrated that predictive training improves performance.

Keywords: Retail Sales Prediction, Ensemble, Machine Learning, RSGDO-RFR, Regression

1. INTRODUCTION

Over the years, the sales pattern is always shifting. In order to prevent shortages or oversupplies, assumptions about supply and demand must be made based on historical data. In order to modify their marketing strategy, some large companies have begun to estimate future sales using previous data. These days, machine learning predictions based on historical data are becoming more and more popular.[1] Information technology is essential for cutting down on data processing time and guaranteeing the quality, consistency, and predictability of results—all of which are crucial for making insightful forecasts and business choices. The retail industry is made up of both organised and unorganised businesses [2]. While organised retail businesses tend to base their judgements on the outcomes of data and information processes, the majority of unorganised retail businesses make decisions intuitively or based on the experience of their owners. An increasing number of unorganised food and grocery markets are being penetrated or occupied by the organised food and grocery industry. Companies in the organised food and grocery sectors are utilising data science for analytics to turn the data into business intelligence for a competitive edge and modern technologies to process data to get better outcomes faster [3].

For the information technology chain store's inventory, management, marketing, customer care, and business financial planning departments, sales forecasting is the most difficult duty. It is extremely challenging to create an effective sales forecasting model for a number of reasons, such as over-forecasting, which raises operating costs and produces unneeded items, and under-forecasting, which reduces customer satisfaction and sales prospects [4-5]. Customer happiness, improved channel connections, and substantial cost savings are all possible outcomes of accurate and reliable sales forecasting. According to Mitchell, one of the technical domains that is expanding the fastest at the moment is machine learning. Data-intensive machine learning techniques are used in a variety of sectors, such as marketing and financial modeling [6]. Additionally, machine learning advances through the discovery of new and appropriate learning algorithms and theories. Additionally, [7] In his article, author demonstrated how machine learning can be used to analyse massive amounts of data in order to address business problems like forecasting future demand and sales. Prediction is a critical component of the corporate value chain. As a result, machine learning solutions for sales prediction are showing promise and are crucial for every organisation as a useful guide. It is also worthwhile to consider identifying the most effective model to deal with the particular issue [8].

In Figure 1, the value of the products sold by Chinese merchants is determined by the National Bureau of Statistics' monthly Retail Sales statistics. Retail sales fluctuations are frequently used as a gauge of consumer spending. With the YoY reading comparing sales values in the reference month

with the same month a year earlier, percentage changes show the pace of changes in such sales. For the Renminbi (CNY), a high reading is typically interpreted as bullish, and a low reading as bearish.

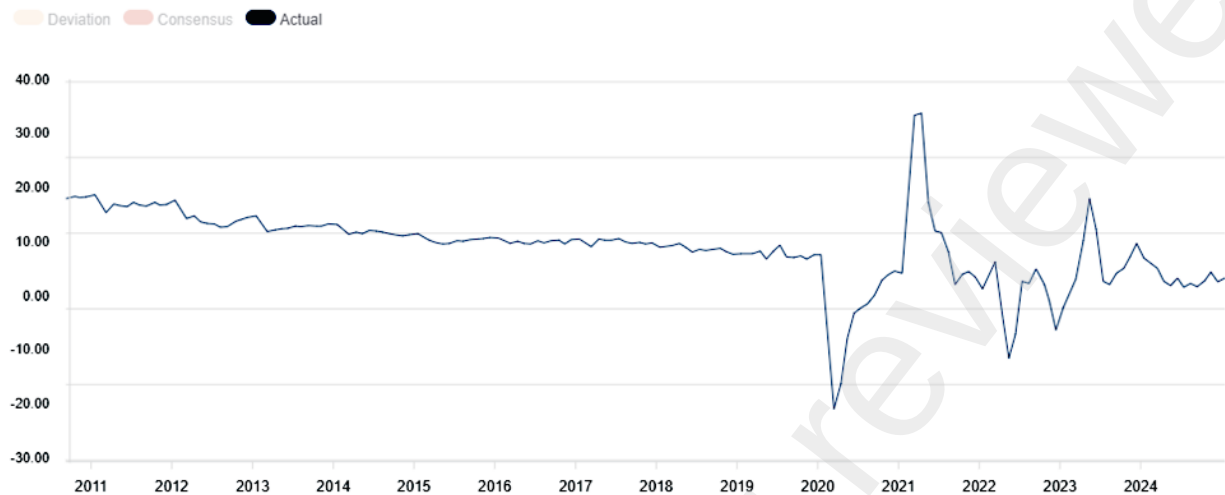


Figure 1: Retail Sales statistics of YoY in china

Accurate sales forecasting is essential for strategic planning, resource allocation, and inventory management in the contemporary retail environment [9]. By guaranteeing product availability, forecasting future sales helps companies optimise stock levels, cut waste, and boost customer happiness. Forecasting has always depended on statistical techniques such as LR [10]. The dynamic and complicated structure of retail sales data, which includes several product families, fluctuating seasonality patterns, and other outside variables, frequently makes these approaches ineffective. As a result, there is increasing interest in using sophisticated hybrid machine learning techniques in finance and other disciplines [11] to improve the precision and dependability of sales projections.

Due to its capacity to analyse data quickly, precisely, and accurately utilising mathematics, statistical and econometric applications, and to provide business information for the creation and implementation of corporate plans, machine learning applications in data science are growing daily [12]. The goal of this project is to use machine learning algorithms to estimate retail sales in an effective manner. Examining cutting-edge technologies, strategies, models, and procedures utilised in organised retail sales prediction is the study's secondary goal. The main goal is to use machine learning algorithms to forecast organised retail sales and compare the outcomes of models that have been put into practice. The study's focus is on utilising machine learning methods to forecast Retail sales. Retailers, decision-makers for retail problems, and policymakers for development, regulation, and control are anticipated to find the study beneficial. The identification of cutting-edge technology for sales prediction in planned retail events with precession organises the job. To

forecast outlet sales, it uses an approach that includes pre-processing the data, developing a prediction model, applying the model to the training data set, determining which model is optimal for prediction, and applying the model to a test set. Lastly, it wraps up the conclusions and ramifications.

2. LITERATURE REVIEW

Table 1: Summary on Related works

Ref	Method	Objective	Result
[13]	Extreme Gradient Boosting (XGBoost)	We provide a machine learning-based sales forecasting methodology that is both effective and precise. In order to extract characteristics from past sales data, feature engineering is first carried out.	uses less memory and computation time and does a very good job of predicting sales.
[14]	Convolutional Neural Networks (CNN)	the CNN model to predict time series data on sales on a weekly and monthly basis. In this study, we compare the performance of the CNN and the FB Prophet technique to determine which is more useful for sales forecasting.	The CNN model outperforms the Facebook Prophet model, which has a weekly predicting value of 307.5.
[15]	Random Forest and decision tree	The suggested system's objective is to give Big Mart Companies a formula for projecting their future sales based on their historical sales.	the prediction take in to an data element with highest prediction of regression methods.
[16]	Machine learning and Deep Learning	To produce more accurate results, the suggested big-mart sales prediction system builds a recurrent multi-level generative adversarial	It demonstrates that the suggested DL technique outperforms other current sales prediction models by 5% to 10%.

		network, deep reinforcement learning engines, and a multivariate poisson distribution scheme.	
[17]	Machine Learning Algorithm	This study examines the possibility of a more precise sales forecast model for companies, emphasising how crucial it is to comprehend consumer requirements and manage effectively in order to preserve a competitive edge.	This study evaluates a number of machine learning models and finds that the model performance has increased as a result of training numerous models predictively.
[18]	ensemble approach DNN + BiLSTM	For the development of an ensemble method to retail sales forecasting, we propose establishing a basic forecaster pool that comprises both individual and pooled forecasting techniques.	The experiment analyzed with other classifiers the suggested algorithm has highest accuracy of 97.94%
[19]	Multiple Linear Regression Models and ARIMA	The paper analyzes China's total retail sales of consumer goods from 2001 to 2020 using various regression models. It finds income, consumption, and tertiary industry factors as significant determinants. The study suggests rationalization for increasing sales and promoting economic development.	Between 2021 and 2030, China's overall retail sales of consumer products will progressively rise.
[20]	Adaptive Genetic Alagorithm (AGA-LSTM) neural network.	The LSTM neural network creation method makes use of the adaptive genetic algorithm	The usefulness of this approach is confirmed by the findings on the Online Retail

		(AGA) to anticipate product categories and sales volume, improve prediction accuracy, and optimise network parameters.	II dataset at UCI, which demonstrate that the AGA-LSTM model's prediction accuracy is significantly higher than that of the conventional LSTM model.
[21]	Bayesian model averaging (BMA)	Using time-series analysis and Bayesian model averaging, this study suggests a three-step data-driven cluster-based demand forecasting method for the retail sector that divides customers into groups according to RFM features.	This paper applicability is proved by a comparative case study analysis, indicating an improvement in the accuracy of daily demand forecast.
[22]	logistic model	Using updated epidemiological data prior to June 16, 2020, the FbProphet and logistic models are utilised to forecast the COVID-19 pandemic trend.	According to our modelling results, an estimated 14.12 million individuals will be infected during a global outbreak that is predicted to peak in late October.

3. METHODOLOGY

Dataset

The Kaggle dataset used in this study comes from the Ecuadorian retail chain Favorita Stores [23]. It includes a number of attributes and metadata that are necessary for forecasting sales. A time series of store numbers, product families, promotional indications, and sales numbers are all included in the training dataset. goods families (family) categorise the kind of goods sold, whilst shop numbers (store_nbr) identify the particular store. Fractional numbers are allowed to accommodate for partial sales (e.g., 1.5 kg of cheese), and the sales statistics (sales) indicate the overall sales for a product family at a specific store on a given day. Promotional indicators (onpromotion) show how many products in a product family are being sold at a store on a given day. Predicting sales for the 15 days after the final date in the training data is the aim of the testing

dataset, which keeps the same properties as the training dataset. A number of supplemental files also include metadata and outside variables that could affect sales. These include oil.csv (daily oil prices), stores.csv (store metadata, including city, state, type, and cluster), and holidays_events.csv (holiday and event details). The additional data facilitates comprehension of outside factors influencing sales, such as changes in the price of oil and the effects of holidays.

Data Pre-processing :

There are several standardised forecasting techniques available. Their relative forecasting performance varies depending on the amount of quantitative complexity employed and the logic foundation utilised to generate the prediction. After converting the data into days, weeks, and years, we checked for outliers and null or missing numbers, and we eliminated them all. To do testing, we cleaned up the dataset. These techniques fall into three categories: qualitative, historical projection, and informal.

Outlier Detection and Removal

Outliers can degrade the efficiency of machine learning models by distorting statistical relationships among features. To eliminate outliers, we employ Z-score evaluation, which determines how far each data point deviates from the mean in standard deviations. A Z-score greater than 3 or less than -3 denotes an outlier that should be eliminated. The Z-score is determined as:

$$Z = \frac{X - \mu}{\sigma} \quad (1)$$

Where X is the data point, μ is the mean of the attribute, and σ is the standard deviation.

Feature Extraction

After the independent variables were finalised, the data was additionally processed using the PCA approach to reduce the high dimensionality of the dataset. In this study, PCA is used to address one of the shortcomings of previous models, which is to minimise data complexity, avoid overfitting, increase computing efficiency, and get rid of redundancy.

For machine learning to be successful, a number of elements are crucial. The performance of machine learning models is greatly impacted by feature selection, which is a crucial component. It decreases training time, increases model accuracy, and prevents overfitting by eliminating redundant data. To solve these issues, we employed a variety of strategies, such as the correlation technique. During the feature selection procedure, feature sets with negative correlations to the

target variables were eliminated.

Principle Component Analysis (PCA)

Principal Component Analysis is a data processing technique for unlabeled extraction of features. Features will be displayed on a newly created, smaller feature space. The features with the most essential data are the new features that were discovered from the PCA extraction findings. By optimizing data variance, the primary constituents are acquired. Data visualization is possible in a low-dimensional principal component space since the number of additional dimensions (features) is less than the number of initial attributes. Calculate the dataset based on each attribute as follows in equation 2,

$$\bar{w}_i = \frac{1}{s} \sum_{j=1}^s w_{ij}, \quad i = 1, 2, \dots, s \quad j = 1, 2, \dots, s \quad (2)$$

Where,

n - Number of features,

w_{ij} - Data j -sample with i -feature,

s - Number data of sample, and

$\bar{w}_i$ - Data i -feature.

Calculate Φ with the following Equation 3,

$$\Phi = [\Phi_{ji}] = [w_{ji} - \bar{w}_i] \quad (3)$$

Φ - Matrix of size $m \times n$.

Calculate the covariance matrix using the following Equation 4,

$$D = \frac{1}{m-1} \Phi^S \Phi \quad (4)$$

D - Matrix of size $n \times n$.

Calculate the features of the D matrix by calculating the subsequent Equation 5.

$$\text{Det}(\lambda J - D) = 0 \quad (5)$$

D - Covariance matrix, and

J - Identity matrix.

After that, calculate the subsequent equation to determine the eigenvectors w that correspond to the features of the record λ ,

$$(\lambda J - D)w = 0 \quad (6)$$

Form matrix w' using the associated features after sorting the eigenvectors according to the eigenvalues, starting with the greatest. Calculate the principal components as follows.

$$PC = \Phi w' \quad (7)$$

Bagging algorithm

Ensemble learning is a technique that combines multiple learners into one strong learner, transforming a single weak learner into an integrated strong learner. This method involves implementing algorithms and combination approaches to improve performance. One modern approach is bagging and boosting, which uses the Bagging method to predict loads for the near future. This technique reduces output variance and enhances the model's generalisability for unstable basic retail sales prediction. Bagging also reduces training time simultaneously. The Bagging method, which relies on resampling from the Bootstrap method, increases the integrated model's generalisability while decreasing variance. Bagging's output remains stable and has some noise-cancelling properties.

Retail Stochastic Gradient Descent Optimization based on Random forest Regression (RSGDO-RFR) Algorithm

The Retail stochastic gradient descent optimisation (RSGDO) algorithm is the most often used optimisation method among machine learning specialists. Academics and industry experts have worked hard to optimise RSGD's runtime performance and provide a theoretical foundation for its empirical success. For instance, it has been demonstrated that RSGD is adequate for training deep neural networks, which has significantly contributed to a large portion of their recent advancements. Here, we provide three pieces that highlight the positive features of RSGD. We start with experimental examples that show how well RSGD works for deep training and emphasise the importance of commencement and acceleration. Next, we use RSGD to explore theoretical relationships between the trainability and generalisation of a model.

In contrast equation 8, every training example $x(j)$ and $y(j)$ undergoes a parameter update via stochastic gradient descent (RSGD):

$$\theta = \theta - \eta \cdot \nabla \theta J(\theta; x(j); y(j)) \quad (8)$$

Batch gradient descent recalculates the gradients for comparable samples prior to altering a parameter, causing repeated computations for large datasets. RSGD removes capital redundancy by

executing updates one at a time. Because RSGD updates quickly and with a lot of variance, the objective function varies a lot. Batch RSGDO approaches the minimum requirement of the corporate governance basin in which the circumstances are formed, but it may also leap to new and perhaps superior local minima. But in the end, this makes convergence to the exact minimum in the estimation more challenging since RSGDO will keep going too far. However, it has been demonstrated that when the bio-tech ratio is gradually reduced, RSGDO exhibits the same convergent behaviour as gradient descent in batches. RSGDO very definitely converges to a local minimum for nonconvex optimisation and to the global minimum for convex optimisation.

The gradient calculation becomes pricey in some situations, such as most machine learning applications, when the objective function has an unreasonably high number of product terms (often a sum over training data points). We'll investigate a suggested way to get around this problem in practice. The idea is simple and embodies a general idea of the creation of the RGDO-RFR algorithm: If a precise calculation is too expensive, substitute a less expensive guess. In particular, as is usually the case, we will investigate the potential of computing an unbiased, low-cost, random estimate of the gradient rather than an accurate one. The family of algorithms that emerge from this randomness (sometimes called stochasticity)-based technique is referred to as RSGDO. As one might expect, such a stochastic method is not free. Since we are trading an accurate quantity (the precise gradient of the objective function) for a low-cost approximation that is susceptible to variation, the final algorithm will have random (erratic) steps rather than clean descent steps. Taking everything into account, this choice about retail sales is highly beneficial for several reasons:

- In terms of economy, RGDO-RFR techniques can finish many more steps in a significantly less amount of time than (precise) gradient descent can finish even one step. Within the allocated computer budget, the precise method might not always be able to finish even one step. By this measure, choosing between an exact and a stochastic approach essentially comes down to choosing between an algorithm that can't start at all and one that, in spite of its instability risk, at least begins
- It has been empirically demonstrated that, even in machine learning contexts, using stochasticity rather than a strong exact approach yields better outcomes, even when the exact algorithm can be run rapidly enough. In machine learning, choosing the right model for the task is conceptually different from but related to minimising the objective function on the training data. By (i) developing models that successfully interpolate the training dataset and (ii) avoiding overfitting, optimisation plays a critical role in machine learning. This allows

the models to generalise well to occurrences that were previously unknown but in-distribution ("similar"). It has been demonstrated that using stochastic gradient descent improves performance on this second goal and decreases overfitting. This is partly due to the fact that noise enables the algorithm to steer clear of saddle points and local minima.

The empirical regret minimization objective $f = J_{emp}$ defined above is a sum of k terms, one for each example in the dataset. When the Store retail sales product dataset is large, evaluating the gradient of J_{emp} at each variable of gradient descent can be computationally expensive. In this case, we can replace at each variable the exact gradient ∇f with a cheap, unbiased estimator $\tilde{\nabla} f$ of it. Equation 9 denoting with $\mathbb{E}t$ the expectation conditioned on all past random choices (that is, all randomization used at times $1, \dots, t-1$), the estimator $\tilde{\nabla} f(it)$ satisfies $\mathbb{E}t [\tilde{\nabla} f(it)] = \nabla f(it)$.

$$it+1 := it - \eta \tilde{\nabla} f(it) \text{ where } \mathbb{E}t [\tilde{\nabla} f(it)] = \nabla f(it). \quad (9)$$

Regarding a scalar function $f : \Omega \rightarrow \mathbb{R}$ examine the general problem of unconstrained optimization as shown in equation 10:

$$\arg \min_{w \in \Omega} f(w) \quad (10)$$

Variable approaches generate a series of solutions $(w_0, w_1, \dots)$ in an attempt to locate a solution. By definition, first-order techniques cannot produce this sequence by taking into account anything other than the iterates and the function's value and gradient at various places in Ω . Therefore, only when f is at least (sub)-differentiable are first order approaches appropriate. Perhaps the easiest and most natural first order method is RSGDO. Since the direction of the fastest reduction of f at any point w is $-\nabla w f$ (the antigradient), RF begins at a randomly selected point $w_0 \in \Omega$ and generates each subsequent point by applying the update in equation 11.

$$w_{t+1} = w_t - \alpha_t \nabla w_t f \quad (11)$$

where $\alpha_t \geq 0$ is a suitable "step size" selection. Occasionally, we shall represent (12) in a more concise manner as the function (update) $G_{f,\alpha} : \Omega \rightarrow \Omega$.

$$G_{f,\alpha}(w) = w - \alpha \nabla w f. \quad (12)$$

Consider Machine Language minimizing an average of functions in Retail sales as equation 13

$$\min_x \frac{1}{m} \sum_{i=1}^m f_i(x) \quad (13)$$

With $\nabla_{\sum_{i=1}^m} f_i(x) = \frac{1}{m} \sum_{i=1}^m \nabla f_i(x)$ gradient descent would repeat:

$$x^{(k)} = x^{(k-1)} - t_k \cdot \frac{1}{m} \sum_{i=1}^m \nabla f_i(x^{(k-1)}), k = 1, 2, 3, \dots \quad (14)$$

Comparatively, the sales forecasting based on product using SGD (or incremental gradient descent) repeats in equation 15 :

$$x^{(k)} = x^{(k-1)} - t_k \cdot \nabla f_{i_k}(x^{(k-1)}), k = 1, 2, 3, \dots \quad (15)$$

when $i_k \in \{1, \dots, n\}$ is variable of CG is k

Two rules to follow the index of i_k at variable k :

- Apply the Randomized rule: evenly select $i_k \in \{1, \dots, n\}$ at random
- Apply the Cyclic rule, selecting $i_k = 1, 2, \dots, n, 1, 2, \dots, n, \dots$
- In practices, rules were Randomised increase prevalent.

From the Randomised cycle is noted as

$$E[\nabla f_{i_k}(x)] = \nabla f(x) \quad (16)$$

For every stage of RSGD may be seen as employing an impartial estimate of the gradient The primary attraction of RSGD It can also result in significant savings for the company in terms of memory use and capital costs because iteration cost is independent of m (number offunctions).

Random Forest: Using several decision trees, Random Forest is a supervised machine learning ensemble approach. It makes use of a method called Bootstrap aggregation, often referred to as bagging, which attempts to simplify models that overfit the training set. Instead of relying on a single decision tree, this method will merge several decision trees to get the final result.

Using several models that have been trained on the same data and averaging their output to get a more potent prediction or classification result is known as ensemble learning. We expect that each model's (in this example, decision tree's) mistakes are independent and vary from tree to tree, which is a prerequisite for ensemble learning. Ensemble machine learning methods provide the following advantages: they can increase prediction accuracy and generate more reliable forecasts; Adaptable to a range of uses, including medical ;Ensemble machine learning algorithms' difficulties may need more computing power, be more difficult to understand than individual models, and need to be carefully chosen.

Bootstrapping: The technique of randomly selecting subsets of a dataset across a predetermined number of variables and iterations is known as bootstrapping. A more potent result is then obtained by averaging these outcomes collectively. One example of an applied ensemble model is bootstrapping.

Random Forest Regression: Several randomly selected decision trees are generated from the data using the bootstrapping Random Forest algorithm, which blends ensemble learning techniques with the decision tree framework. The outputs are then averaged to provide a new result that frequently produces accurate predictions or classifications.

The original purpose of Random Forests Regression (RFR) was to handle large datasets with strong statistical effectiveness. Because of its exceptional accuracy, this approach excels in prediction tasks. Regression, classification, and other learning-related tasks are all handled by RF, which functions as a classic machine learning algorithm. After utilising the bagging technique to aggregate data from the original dataset, the decision tree model is used to train each aggregated group separately. The decision results from these sub-models are combined and analysed to create the final RF model. Voting is the basis for the final RF model's operation; the classification with the most votes becomes the final output. This method improves the overall classification accuracy by reducing the error of individual classifiers. Notably, RF performs better in terms of accuracy and classification performance than competing techniques like regression trees and neural networks. Its effectiveness is especially noticeable while processing massive amounts of data.

Here are some of the crucial factors that are highlighted:

- **n_estimators** – The quantity of decision trees you plan to execute within the model
- **criterion** – This variable lets you choose the loss function, or criteria, that will be used to decide the model's results. We have the option to choose among loss functions like mean absolute error (MAE) and mean squared error (MSE). MSE is the default value.
- **max_depth** – This establishes each tree's maximum depth.
- **max_features** – The most features that the model will take into account when deciding on a split
- **bootstrap** – Since the default answer for this is True, the model adheres to the previously stated bootstrapping criteria.
- **max_samples** – This option doesn't apply until bootstrapping is set to True, which is assumed. When True, this parameter establishes the maximum sample size for every tree.

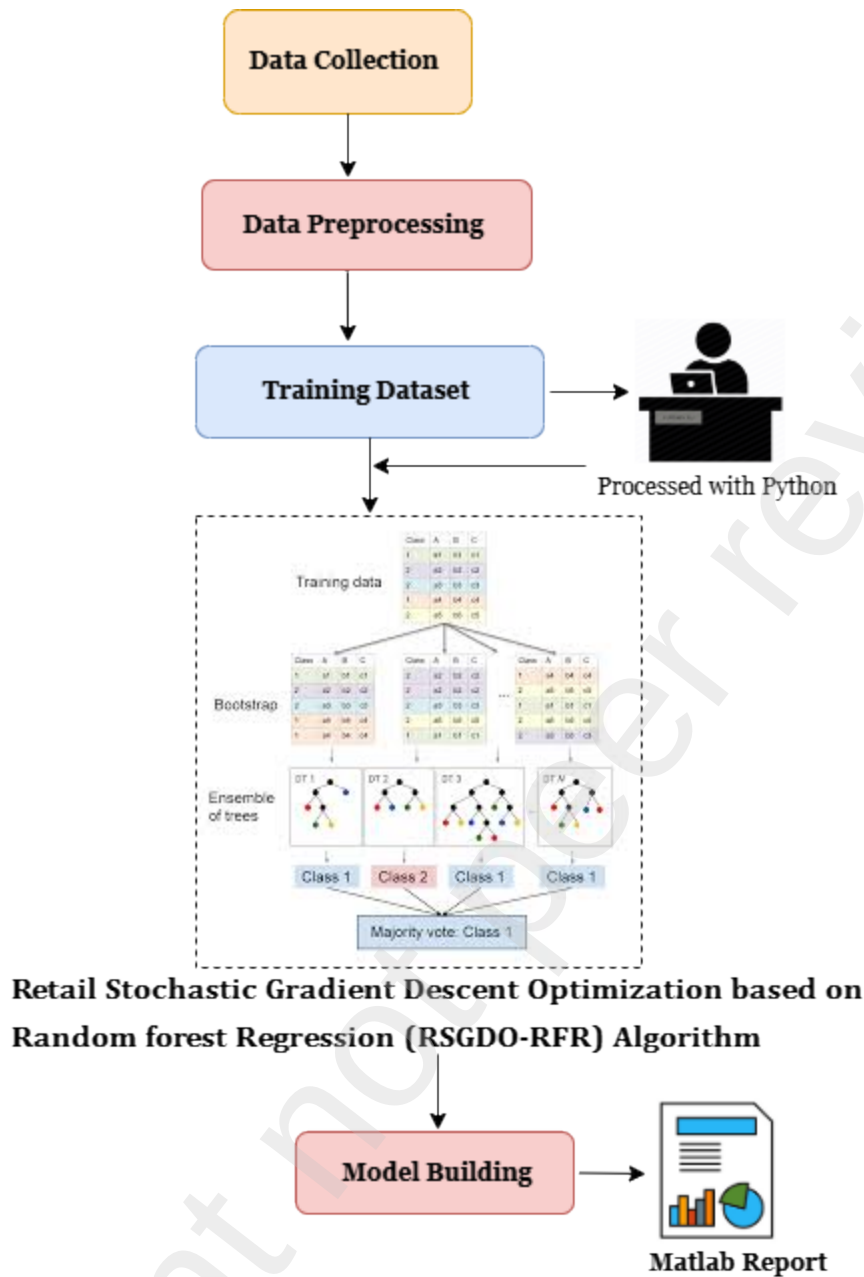


Figure 2 : Proposed Model

In order to identify a more ideal set of parameters, we will first demonstrate some fundamental values that were entered into the random forest regression model. Grid search and cross validation will then be used. Metrics like accuracy allow us to assess how well we performed when we solved categorisation tasks. We may utilise variables like mean squared error, root mean squared error, R2, adjusted R2, and others to examine a regression model's performance indicators.

Algorithm 1: Pseudo Code for RSGDO-RFR Algorithm
<i>Required:</i> α: Stepsize <i>Required:</i> $\beta_x, \beta_y \in [0, 1]$: Inverse capital structure rate of Retail sales approximates <i>Required:</i> $f(\theta)$: Stochastics goal variable with specifications θ <i>Required:</i> θ_0: starting specifications variable $m_v \leftarrow 0$ (Beings with first variable) $v_v \leftarrow 0$ (Beings with second variable) $s \leftarrow 0$ (Beings with delayedstep) While θ_t wrong do $s \leftarrow s + 1$ $g_t \leftarrow \nabla \theta_{f_t}(\theta_{t-1})$ (Gets RSGDO-RFR aim at delayed-step as s) $m_v \leftarrow \beta_x \cdot m_v - 1 + (1 - \beta_x) \cdot g_t$ (First updated biased approximates) $v_v \leftarrow \beta_y \cdot v_v - 1 + (1 - \beta_y) \cdot g_t$ (Second updated biased approximates) $m_{bt} \leftarrow m_v / (1 - \beta_{tx})$ (Initial calculate bias-corrected approximates) $v_{bt} \leftarrow v_v / (1 - \beta_{ty})$ (Secondly calculate bias-corrected approximates) $\theta_v \leftarrow \theta_v - 1 - \alpha \cdot m_{bt} / (\vee v_{bt+})$ (Modify Specification) End While return θ_t (Specification) Result

4. RESULTS AND DISCUSSION

Table 2: Experimental Setup

HARDWARE REQUIREMENTS	SOFTWARE REQUIREMENTS
System : i3 Processor Hard Disk : 500 GB. Monitor : 15"LED Ram : 4GB 5.2	Operating system : Windows 7 or above, linux. Scripting Tool: Jupyter Notebook, Google colab Language: Python3.0

4.1 China's retail sales

The total sales of retail products and services during a certain month are compared to the same month last year to determine the year-over-year change in retail sales in China. In figure 3, from December 2024, China's retail sales increased 3.6% year over year, surpassing the market's forecast of 3.4% and rebounding from a 3-month low of 3.0% in November. The following categories saw increases in sales: food, oil, and grain (9.8% vs. 10.0% in November); tobacco and alcohol (10.3% vs. -3.0%); apparel, shoes, hats, and textiles (0.7% vs. -4.4%); cosmetics (0.7% vs. -26.3%); personal care (6.2% vs. 1.2%); sports and entertainment products (16.6% vs. 3.4%); office supplies (9.0% vs.). Conversely, sales of jewellery (-1.0% vs. -5.6%), petroleum products (-2.7% vs.

-7.0%), drinks (-8.4% vs. -4.2%), and Chinese and Western medicine (-0.8% vs. -2.6%) all saw larger declines. Retail turnover increased by 3.4% during the course of the year. Food, clothes, and personal care sales grew by 6.4% in 2024, accounting for 26.7% of the overall retail trade of consumer products, while online retail turnover climbed by 7.1%.

CN Retail Sales YoY - percent

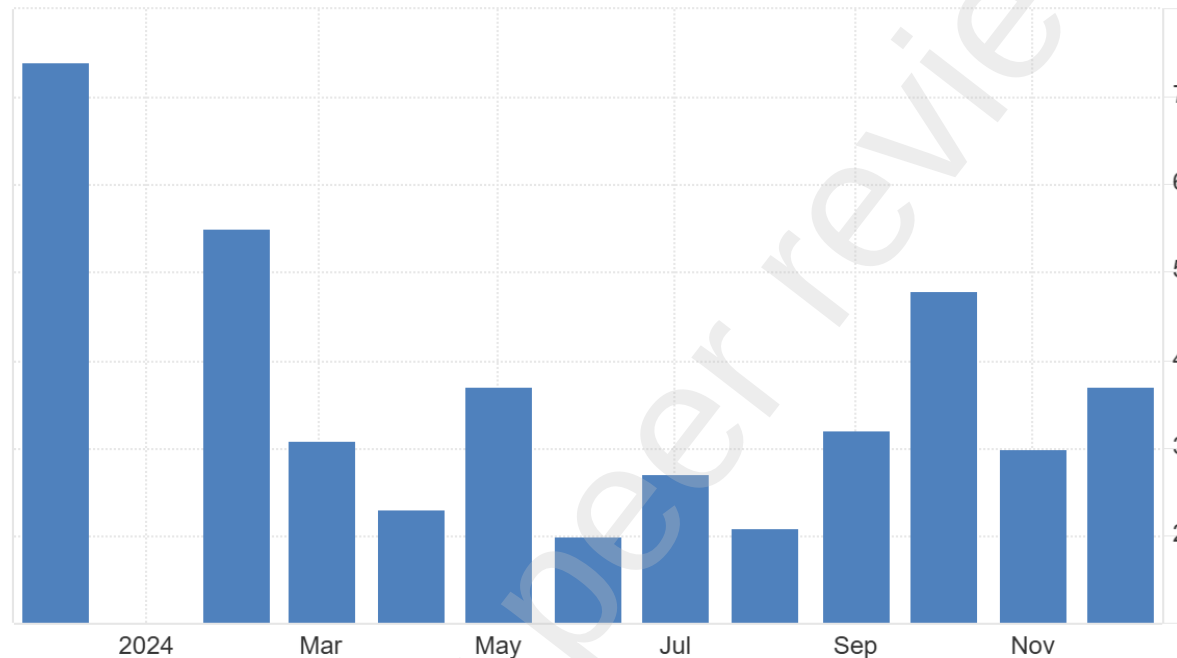


Figure 3: Retail sales in china

4.2 Retail Sales prediction of Consumer Goods in China from 2024 to 2028

Between 2013 and 2023, China's retail sales of consumer products consistently increased, going from 232.24 to 473.10 billion Yuan Renminbi. The yearly percentage increases have been gradually slowing down, though, as seen by the year-over-year growth rates tapering down. For instance, the annual variation decreased to 4.26% in 2023 from 13.00% in 2013. Similar declines have also been seen in the CAGR, which went from 15.90% in 2013 to 4.5% in 2023. Retail sales are predicted to keep increasing in the future, reaching 567.06 hundred billion Yuan by 2028. In Figure 4, the five-year predicted CAGR from 2024 to 2028 is anticipated to be a relatively moderate 2.86% in spite of this development. This suggests that retail sales will rise gradually but steadily over the ensuing years.

Future trends to keep an eye on include:

- E-commerce as a persistently important factor in the expansion of retail sales.
- The trend of consumers purchasing high-end, superior goods.
- How government policies affect consumer spending and confidence.
- How changes in technology affect consumers' purchasing habits.

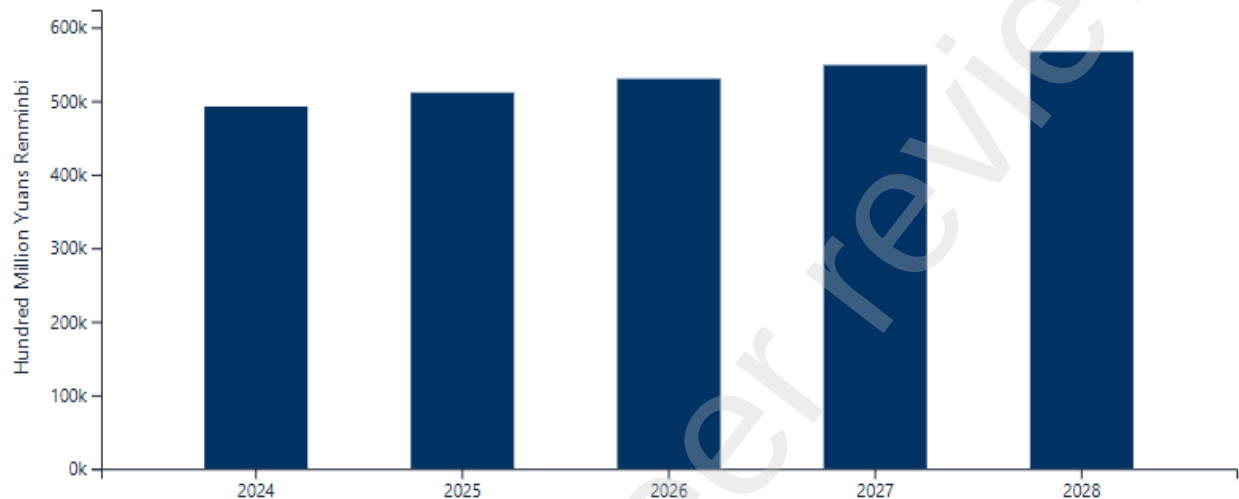


Figure 4: Prediction based on National Bureau of Statistics of china data

4.3 Performance Evaluation

A comparative analysis is carried out between the suggested approach RSGDO-RFR and other well-known machine learning techniques in order to verify its efficacy. Among the techniques are Support Vector Machine (SVM)[24], XGBoost[25], and Traditional Random Forset(RF)[26]. The test results of the dataset approaches with classifiers are summarized in Tables 2 correspondingly. The suggested model is evaluated using accuracy rate of algorithm. One of the most crucial performance indicators for categorization is accuracy.

➤ Accuracy of the classification method

Table 3: Outcome value of Accuracy

Classification Method	Accuracy (%)
SVM	80.1
XGBoost	88.5

RSGDO-RFR [Proposed]	90.7
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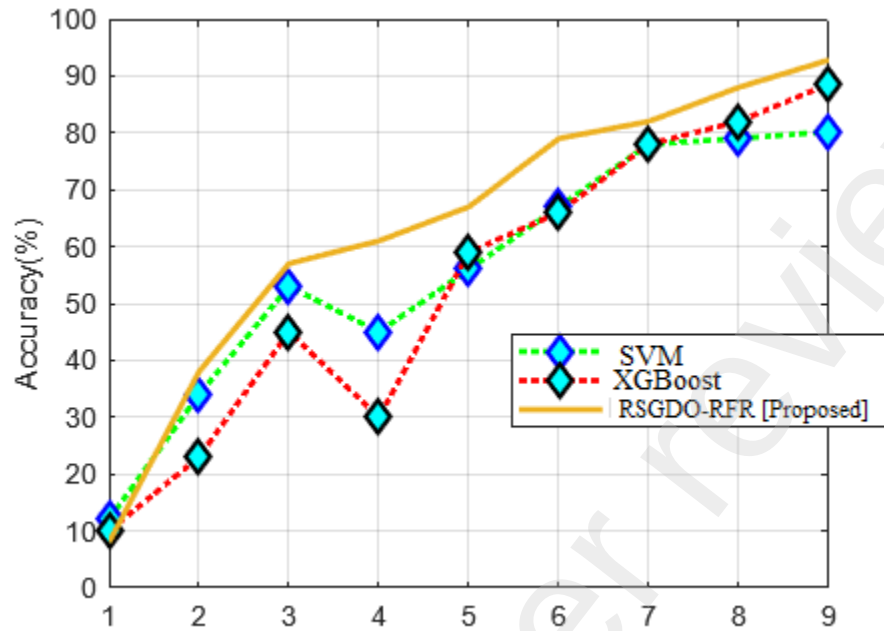


Figure 5: Outcome of Accuracy

The percentage of incidents that are reliably and effectively classified is analyzed by accuracy. Table 3 and Figure 5 show the accuracy's outcome. Our suggested approach had a higher RSGDO-RFR(90.7%) percentage than the current methods, which are SVM (80.1%) and XGBoost (88.5%).When compared to existing approaches, the proposed approach, has significantly improved prediction in Retail sales Prediction using ML Algorithm

4.4 Error Metrics

A number of important measures are used to assess the suggested model's efficacy in Table 4 . The RMSE, which calculates the square root of the average squared discrepancies between the expected and actual sales numbers, is the main statistic. The predicted accuracy of the model is clearly indicated by this statistic. Furthermore, without taking into account their direction, the MAE is utilised to evaluate the average size of prediction errors. To assess the model's overall explanatory ability, the R^2 score and Adjusted R^2 —which shows the percentage of variation in the dependent variable that can be predicted from the independent variables—is also computed. Together, these measurements offer a thorough knowledge of the model's performance, pointing out both its advantages and disadvantages.

Table 4: Outcome of RMSE,MAE, R^2 ,Adjusted R^2

Methods	RMSE	MAE	R^2	Adjusted R^2
SVM	283.56	155.05	0.90	0.91
XGBoost	282.26	140.12	0.92	0.92
RF	255.05	133.31	0.93	0.94
RSGDO-RFR	230.52	100.54	0.94	0.95



Figure 6: Outcome result of RSME and MAE

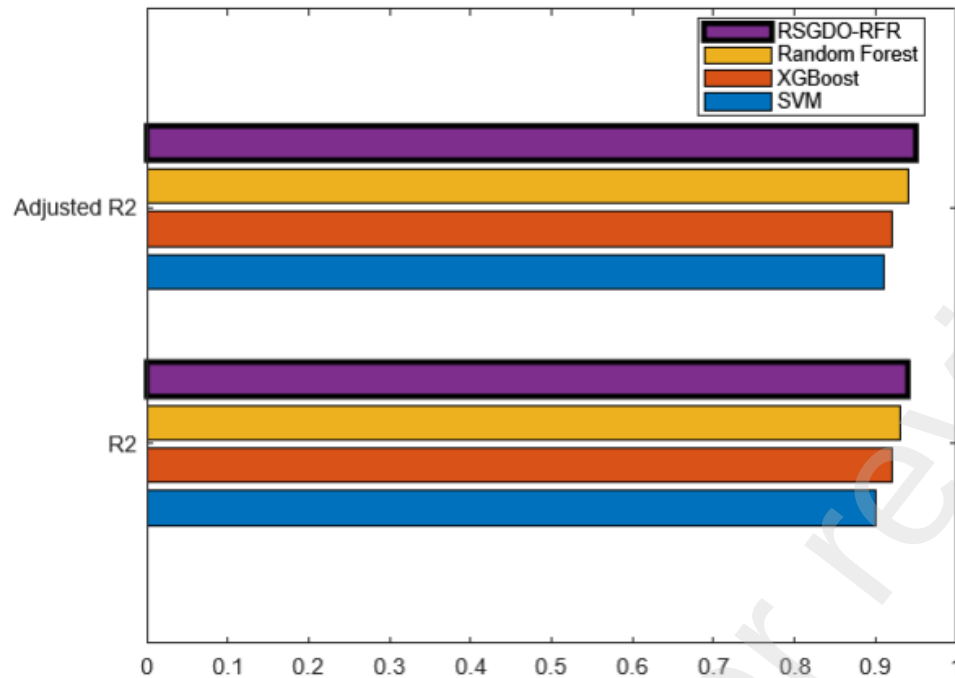


Figure 7: Outcome result of R^2 and Adjusted R^2

Table 4 shows how well the suggested method RSGDO-RFR model performs in comparison to the XGBoost, Support Vector machine (SVM), and Random Forest models. With an R-squared of 0.94, which is 0.3% to 0.5% higher than the others, the Suggested model performs better than the others in practically every statistic. A similar pattern is shown by the Adjusted R squared, which suggests a steady performance. At the RSGDO-RFR model's MSE is the lowest and improves by around 0.5% to 1.8%. Likewise, the Mean Absolute Error and Root Mean Squared are somewhat superior. The RF model better represents the dataset's variability and trend, as seen by the RMSE value of 230.52, which also shows a more improvement.

5. CONCLUSION

Applications of machine learning are being utilised more and more in corporate analytics and data processing. This study shows that compared to previous models, machine learning-based algorithms yield more exact, accurate, and trustworthy decision-making results. The findings can assist managers create competitive marketing strategies, help regulators make the right estimates and policies, and increase merchants' trust in using machine learning in their data processing and analysis. For huge datasets, techniques like parallel processing and distributed computing are employed to increase the effectiveness of model training and prediction. Data improvement techniques are used to increase the stability and generalisation capacity of small datasets. In order to speed up training and enhance prediction, machine learning may also be used to transfer model

parameters that have been learnt on big datasets to smaller ones. Based on past sales data, the study uses three models—SVM, XGBoost, and Random Forest—to predict future sales for a shop. Compared to other methods, the stock strategy produces more accurate forecasts by efficiently combining the model outputs. However, the capacity to mine more information is limited by language limitations and a lack of trials; time series models such as ARIMA may be used in future study.

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Reference

- [1] C. Sekhara, "Retail Sales Prediction Using Machine Learning Algorithms," *Turkish Online Journal of Qualitative Inquiry*, vol. 12, no. 1, pp. 315–322, Sep. 2024, Available: <https://tojqi.net/index.php/journal/article/view/8277>. [Accessed: Feb. 05, 2025]
- [2] B. Zhang, M.-L. Tseng, L. Qi, Y. Guo, and C.-H. Wang, "A comparative online sales forecasting analysis: Data mining techniques," *Computers & Industrial Engineering*, vol. 176, p. 108935, Feb. 2023, doi: <https://doi.org/10.1016/j.cie.2022.108935>. Available: <https://www.sciencedirect.com/science/article/pii/S0360835222009238>
- [3] S. Bhattacharya, S. Chandra, and D. Dhar, "Customer behaviour forecasting using predictive analysis of sales data," *Nucleation and Atmospheric Aerosols*, Jan. 2023, doi: <https://doi.org/10.1063/5.0166610>. Available: <https://pubs.aip.org/aip/acp/article/2876/1/020008/2908830>. [Accessed: Sep. 18, 2023]
- [4] D. Li, F. Gu, X. Li, R. Du, D. Chen, and A. Madden, "Dynamic sales prediction with auto-learning and elastic-adjustment mechanism for inventory optimization," *Information Systems*, vol. 119, pp. 102259–102259, Oct. 2023, doi: <https://doi.org/10.1016/j.is.2023.102259>
- [5] Wang Minfei, Tong Shiyuan, P. Qi, Ge Jianmin, L. Ying, and S. Jian, "Sales Forecasting Model Based on Ensemble Learning and Its Application in Anomaly Detection," Feb. 2023, doi: <https://doi.org/10.1145/3585542.3585566>
- [6] G. Duan and J. Dong, "Construction of Ensemble Learning Model for Home Appliance Demand Forecasting," *Applied Sciences*, vol. 14, no. 17, pp. 7658–7658, Aug. 2024, doi: <https://doi.org/10.3390/app14177658>. Available: <https://www.mdpi.com/2076-3417/14/17/7658>. [Accessed: Nov. 29, 2024]

- [7] C. Liu, W. Xie, W.-Z. Wu, and H. Zhu, "Predicting Chinese total retail sales of consumer goods by employing an extended discrete grey polynomial model," *Engineering Applications of Artificial Intelligence*, vol. 102, p. 104261, Jun. 2021, doi: <https://doi.org/10.1016/j.engappai.2021.104261>
- [8] Z. Huo, "Sales Prediction based on Machine Learning," *IEEE Xplore*, Mar. 01, 2021. doi: <https://doi.org/10.1109/ECIT52743.2021.00093>. Available: <https://ieeexplore.ieee.org/document/9406916>
- [9] "Enhancing Retail Sales Forecasting with Optimized Machine Learning Models," *Arxiv.org*, 2015. Available: <https://arxiv.org/html/2410.13773v2>. [Accessed: Feb. 05, 2025]
- [10] J. Liu, L. Chen, R. Luo, and J. Zhu, "A combination model based on multi-angle feature extraction and sentiment analysis: Application to EVs sales forecasting," *Expert systems with applications*, vol. 224, pp. 119986–119986, Aug. 2023, doi: <https://doi.org/10.1016/j.eswa.2023.119986>
- [11] R. Hasan, M. A. Kabir, R. A. Shuvro, and P. Das, "A Comparative Study on Forecasting of Retail Sales," Mar. 2022, doi: <https://doi.org/10.48550/arxiv.2203.06848>
- [12] S. Raizada and J. R. Saini, "Comparative Analysis of Supervised Machine Learning Techniques for Sales Forecasting," *International Journal of Advanced Computer Science and Applications*, vol. 12, no. 11, 2021, doi: <https://doi.org/10.14569/ijacsa.2021.0121112>
- [13] X. dairu and Z. Shilong, "Machine Learning Model for Sales Forecasting by Using XGBoost," *IEEE Xplore*, 2021. doi: <https://doi.org/10.1109/ICCECE51280.2021.9342304>. Available: <https://ieeexplore.ieee.org/abstract/document/9342304>
- [14] W. Khairul, None Afiqah Bazlla Md Soom, A. Mat, None Juhaida Ismail, None Aszila Asmat, and A. Rahm, "Sales Forecasting Using Convolution Neural Network," vol. 30, no. 3, pp. 290–301, May 2023, doi: <https://doi.org/10.37934/araset.30.3.290301>
- [15] Ryali Ajay, R. Sabin Joel, and G. Om, "Analyzing and Predicting the Sales Forecasting using Modified Random Forest and Decision Tree Algorithm," Jun. 2023, doi: <https://doi.org/10.1109/icces57224.2023.10192723>
- [16] T. K. Thivakaran and M. Ramesh, "Sales Data Analysis and Prediction System for Big Mart using Deep Recurrent Reinforcement Principles," *IEEE Xplore*, Sep. 01, 2022. doi: <https://doi.org/10.1109/ICIRCA54612.2022.9985752>. Available: <https://ieeexplore.ieee.org/abstract/document/9985752>. [Accessed: May 01, 2023]
- [17] T. Kamble, Madhuri Ghuge, R. Rana, Harsh Vardhan, Y. Shelar, and Tushar Machale, "Ensemble Machine Learning Models to Forecast Sales," Dec. 2023, doi: <https://doi.org/10.1109/icimia60377.2023.10426565>
- [18] S. Babu and M. M. Kodabagi, "An Efficient Framework for Predicting Future Retail Sales Using Ensemble DNN-BiLSTM Technique," *SN Computer Science*, vol. 5, no. 1, Jan. 2024, doi: <https://doi.org/10.1007/s42979-023-02427-3>
- [19] Y. Zhang, "Analysis And Prediction of Influencing Factors of Total Retail Sales of Consumer Goods in China——Based on Multiple Linear Regression Models and ARIMA Models," *BCP Business & Management*, vol. 23, pp. 1036–1048, Aug. 2022, doi: <https://doi.org/10.54691/bcpbm.v23i.1496>

- [20] K. Chen, "An Online Retail Prediction Model Based on AGA-LSTM Neural Network," *IEEE Xplore*, Oct. 01, 2020. doi: <https://doi.org/10.1109/MLBDBI51377.2020.00032>. Available: <https://ieeexplore.ieee.org/document/9360965>
- [21] M. Seyedan, F. Mafakheri, and C. Wang, "Cluster-based demand forecasting using Bayesian model averaging: An ensemble learning approach," *Decision Analytics Journal*, p. 100033, Mar. 2022, doi: <https://doi.org/10.1016/j.dajour.2022.100033>
- [22] P. Wang, X. Zheng, J. Li, and B. Zhu, "Prediction of epidemic trends in COVID-19 with logistic model and machine learning technics," *Chaos, Solitons & Fractals*, vol. 139, p. 110058, Oct. 2020, doi: <https://doi.org/10.1016/j.chaos.2020.110058>. Available: <https://reader.elsevier.com/reader/sd/pii/S0960077920304550?token=6C51F1BC3363E096C28C771BB5CE2793BD118C8E6A7C24EE40F8AEFFAE579BB00CA6BF66BDD07D6300796FD5A93B65AC>
- [23] "Store Sales - Time Series Forecasting," *kaggle.com*. Available: <https://www.kaggle.com/competitions/store-sales-time-series-forecasting>
- [24] E Padma, "Intelligent Sales Prediction Using SVM Algorithm," *International Scientific Journal of Engineering and Management*, vol. 02, no. 03, Mar. 2023, doi: <https://doi.org/10.55041/isjem00209>
- [25] K. Li, "A Sales Prediction Method Based on XGBoost Algorithm Model," *BCP Business & Management*, vol. 36, pp. 367–371, Jan. 2023, doi: <https://doi.org/10.54691/bcpbm.v36i.3487>
- [26] S. Pang, "Retail Sales Forecast Based on Machine Learning Methods," *2021 5th Annual International Conference on Data Science and Business Analytics (ICDSBA)*, pp. 357–361, Oct. 2022, doi: <https://doi.org/10.1109/icdsba57203.2022.00030>